The Burden of Burns: An Analysis of Public Health Measures

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Accurate analysis of injuries is paramount when allocating resources for prevention, research, education, and legislation. As burn mortality has improved over recent decades, the societal burden of burn injuries has grown ambiguous to the public while a scarcity of investigational funding for survivors has led to a gap in understanding lifelong sequela. We aim to compare national references reporting the incidence of burn injuries in the United States. The American Burn Association Burn Injury Summary Report (ABA-BISR), American Burn Association Fact Sheet, Centers for Disease Control and Prevention (CDC) Web-based Injury Statistics Query and Reporting (WISQARS) database, the CDC National Center for Health Statistics' National Hospital Ambulatory Medical Care Survey (NHAMCS), National Inpatient Sample (NIS), National Emergency Department Sample (NEDS), and commercially available claims databases were queried for 2020 or the most recent data available. The BISR estimated 30,135 burn admissions in 2022. The 2016 ABA Fact Sheet reported 486,000 burns presented to US emergency departments (ED). In 2020, CDC's WISQARS database reported 3,529 fatal, and 287,926 non-fatal, burn injuries. The 2020 NEDS reported 438,185 ED visits while the 2020 NIS estimated 103,235 inpatients. The NHAMCS reported 359,000 ED visits for burn injuries in the same period, and an analysis of ICD-10 burn codes demonstrated over 698,555 claims. Our study demonstrates a large variability in the reported incidence of burn injury by the ABA, CDC, national samples, and claims databases. Per our analyses, we estimate that 600,000 individuals annually suffer a burn injury which merits emergent care in the United States.

Key words: burn injury; incidence; advocacy; estimate; burden; burns.

INTRODUCTION

An accurate estimate of the magnitude and scope of injuries and illnesses is paramount for resource allocation, public policy creation, population health surveillance, and research. Traditionally, incidence measurements have been conducted through quantitative epidemiologic analyses using systematic population-based surveys.[1,2] Unfortunately, these data are often incomplete or otherwise inadequate for usefulness. Whether due to challenges with interfaces, data formatting, or data reporting, epidemiological and public health analyses possess variability due to differences in the reporting sources.[3] Quantifying the degree of this variability is all the more important because studies over multiple decades have found that the National Institutes of Health (NIH) funding levels are highly associated with burden of disease.[4–7]

Burn injuries are treated across a wide range of facilities, from rural emergency rooms to comprehensive urban burn centers (BCs). BCs are designed to deliver specialized burn care to burn patients and are staffed by expert multidisciplinary teams, processes, and facilities designated to optimize treatment and rehabilitation. Of the approximately 6200 hospitals in the United States, only 135 have a burn center, and of these, only 50%-60% are verified by the American Burn Association (ABA).[8,9] The growing number of hospitals in the United States is expected to be a driver of new BCs.[10,11] The US BC market size was valued at $5.9B in 2021 and is expected to expand at a compound annual growth rate (CAGR) of 3.6%

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Conflict of Interest: Dr. Phelan serves as a consultant with Avita Medical and SpectralMD. Dr. Carter is a stockholder for SpectralMD and serves as a Medical Consultant for SpectralMD. He also serves as a consultant for Avita Medical and Polynovo Ltd. where, in lieu of compensation, all proceeds are donated to a local burn charity to support burn education, survivor programs, outreach, and research. E. Murata is employed by BData Inc. B. Phillips serves as the Chief Executive Officer of BData Inc. A. Ivanko, Dr. Garbuzov, Dr. Schoen, R. Kearns, and D. Danos have nothing to disclose.

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Table 1. Sources of Burn Injury Incidence Reporting

	ABA BISR	ABA Fact Sheet	CDC NHAMCS	NIS	NEDS	CDC WISQARS	Claims data
Patient population	Admissions	ED & admissions	ED	In-patients	ED	Unintentional injuries	Claims
Burn injury estimate	30,135	486,000	359,000	118,720	438,185	287,926 non-fatal 3,529 fatal	698,555
Estimate year	2022	2011	2020	2020	2020	2020	2020

from 2022 to 2030.[10] However, currently the majority of burn care occurs at rural, non-trauma, non-burn centers. This makes it challenging to understand the patient population and quantify their care. Further, while the volume of burn injuries has remained the same over the past few decades, burn mortality has improved with the increase in burn centers, leading to an unclear societal burden of burn injuries to the public.

We sought to compare national references which report the incidence of burn injury and its related sequela in the United States and to use those signals to inform an accurate estimate of the burden of burns in the United States.

METHODS

The ABA Burn Injury Summary Report (BISR), ABA Fact Sheet, Centers for Disease Control and Prevention's Web-based Injury Statistics Query and Reporting (CDC-WISQARS) database, the CDC National Center for Health Statistics' National Hospital Ambulatory Medical Care Survey (NHAMCS), National Inpatient Sample (NIS), National Emergency Department Sample (NEDS), and commercially available claims databases were queried for 2020 or the most recent data available. Non-US reports, claims, or analyses were excluded from our review. The costs of burn injury were reported in US dollars where available. Claims analyses were conducted using ICD-10 burn injury codes from a commercially available database incorporating managed care, Medicare, and Medicaid payors.

RESULTS

The ABA BSIR gives an up-to-date annual burn injury admission estimate; however, the data come from only 109 BCs, thus only providing information for the BCs submitting data onto the platform. It estimated 30,135 burn injury admissions in 2022, a fall from the 32,857 admissions estimate in 2021. The ABA Burn Incidence fact sheet, created in 2015,[12] acknowledges that some burns may be treated at locations other than a burn center. Data displayed in the fact sheet account for both ED visits and admissions, providing a greater incidence of burn injury. The data are based on the 2011 National Hospital Ambulatory Medical Care Survey (NHAMCS) data and are funneled from the US Census Bureau, through the CDC. The 2011 NHAMCS data come from an ambulatory survey, and estimate emergency room (ER) visits for burns of only 419 emergency departments (EDs), and of those, only 387 responded. Interestingly, the NHAMCS surveys each hospital over a 4-week period and generalizes it to an entire year. The 2020 NHAMCS data report 359,000 ED visits relating to burn injury.[13] A total of 422 EDs were identified for this survey, and 204 of which fully responded. Given that the NHAMCS surveys solely ERs, it is highly likely that much inpatient and outpatient data are missing.

The most recent publicly available NIS and National Emergency Department (NED) data, published in 2016, include data from 49 US states, and report 53,220 inpatient stays for burn injury in 2013, and 409,183 ER visits in that same year.[14] The 2019 NIS data report 90,465 burn inpatients annually, the most recent publicly available NED Statistics Fact Sheet from 2013 reports 409,183 burn injury-related ER visits.[14] The 2020 NIS data estimate 89,555 inpatient stays with burn diagnosis, and 29,165 inpatient stays with a burn-related diagnosis related group (DRG) code, indicating an estimated 118,720 total inpatients for that year. The 2020 NEDS estimates 438,185 patients in the ED with burn-related diagnosis codes in 2020.

In 2010, the CDC's Web-based Injury Statistics Query and Reporting (WISQARS) database estimated 412,256 burn injury-related ED visits. The cost of medical care is estimated to be approximately $1.5B and another nearly $5B in costs associated with lost work in 2010 for burn injury.[15] In 2020, CDC WISQARS reports a total of 291,455 burn injuries, with the respective costs to be $4.1B for medical costs, and $721M in loss of work, associated with burn injury.[15] Although we see a decline in the decade span of burn injuries through this database, as mentioned previously, the CDC NHAMCS reported 359,000 injuries during that same time. The $3.1B increase in medical costs may be attributed to an increase in burn patients across the United States. The data demonstrate the high likelihood that these reports do not encompass burn injuries treated at rural hospitals, in the inpatient setting, or in the outpatient setting. Over the same time, 698,555 insurance claims regarding burn injuries have been filed. These are likely underreporting self-pay/uninsured, military facilities, and endowment or community-based hospitals. We are not able to distinguish admissions and potential sequelae that may be in this dataset.

Table 1 summarizes the 7 sources of burn injury reported which we were able to identify. Per our analyses, we estimate that approximately 600,000 individuals per year obtain burn injury emergent care in the United States.

DISCUSSION

Our study demonstrates a large variability in the reported incidence of burn injury by the ABA, CDC, national samples, and claims databases. Given that incidence data reports capture admitted burn patients and those seen in the ED, outpatient facility data and long-term complications are not indicated by these reports. The unknown true incidence, and thereby the actual cost, of burn injuries has led to a scarcity of investigational funding for burn patients and survivors, leading to a gap in our understanding of lifelong sequela.

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We estimate that 600,000 individuals annually suffer a burn injury which merits emergent care in the United States. On average, patients stay in the intensive care unit (ICU) for one-half to one full day per percent TBSA burned. An overall decline in mortality has been reported around the world.[16] With advancements in overall burn care survival has become an expected outcome, shifting focus on optimizing long-term management of these patients. Typically, postdischarge, burn patients are seen every week until all wounds are closed, then range from 1-month to 3- to 6-month follow-up visits for at least a year. When patients experience sequelae postburn injuries such as psychiatric conditions, hypertrophic scaring, range of motion disabilities, etc. they are seen even longer by the burn center providers. Long-term population studies have linked burn injury with an elevated risk of cancer, cardiovascular disease, neurological disorders, diabetes, musculoskeletal disorders, gastrointestinal disease, and infections.[17,18] Burn patients also suffer from nutrient deficiencies, a situation complicated by the fact that most insurance companies will not cover simple items such as multivitamins. Patients face the consequences of burn hypermetabolism for years following their initial injury. Unfortunately, these sequelae are unaccounted for in-burn incidence measurements. Accurate medical coding becomes crucial in ensuring providers have a comprehensive understanding of a patient's history, enabling the delivery of proper and personalized care, to better patient outcomes. In addition, appropriate codes play a pivotal role in insurance companies determining coverage, reimbursement, and claim settlements. In all, this process identifies areas with high healthcare utilization, facilitating targeted interventions and resource utilization.[19]

In the healthcare systems, there are currently no DRG codes that allow healthcare providers to associate long-term complications to the burn injury ICD-10 code. By not accounting for these sequelae, the currently available signals for burn injury underestimate the scope of the problem which in turn leads to a scarcity of funding from the NIH, which currently funds 315 research and disease areas.[20] To put this into perspective, the 2020 estimated incidence of Alzheimer's as indicated by the NIH funding report equates to 500,000 cases and 175,00 deaths, resulting in a total of $2.9B being allocated for research funding that year.[20] Arguably, many illnesses have a similar incidence and sequelae rate as burn injury; however, the lack of reporting of the true estimate of burns places patients and providers at a disadvantage when it comes time for resource allocation.

The nation's healthcare agenda is complex, with a considerable number of healthcare issues having an impact on providers worldwide. With multiple stakeholders competing for limited resources, the need for engagement and advocacy has never been greater. Advocacy efforts play a crucial role in public policy creation, and guiding resource allocation. In the healthcare setting, unsuccessful advocacy results in increased complications and death.[21] A proper estimate of the burden of burn injury provides a comprehensive understanding of the disease and will allow for funding to optimize patient care.

CONCLUSION

There is large variability in the reported incidence of burn injury by the ABA, CDC, national samples, and claims databases. The true incidence of burn injuries is likely much higher than reported by the American Burn Association, approximately 600,000, suggesting that burn care costs are much higher than previously reported. The unknown actual cost and incidence of burn injuries have led to a scarcity of investigational funding for burn patients and survivors, leading to a gap in our understanding of lifelong sequela. Further analysis of claims databases is needed to better determine treatment patterns.

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